Term Paper

Geopolitical Risk and the Structural Fragmentation of Global Trade Architecture

Intermediate Econometrics

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Executive Summary

International trade today features highly integrated and robust supply chains. Even so, modern international trade faces systemic changes that can significantly impact the operations of entire countries. This report examines the extent to which geopolitical risk, considered as a shock factor, contracts aggregate trade openness at the national level. It develops an index defined as the sum of exports and imports as a percentage of GDP, isolating historical and structural relationships, as well as path dependencies (Mulabdic & Yotov, 2025). A longitudinal panel dataset from 2010 to 2024 is used, and the empirical analysis evaluates 3 nested panel specifications:

- Pooled Ordinary Least Squares (OLS)

- Two-Way Fixed Effects (TWFE)

- Dynamic Panel Data framework.

The Two-Way Fixed Effects statistic, which is the recommended statistic, shows that for each unit increase in the Geopolitical Risk (GPR) index, trade openness will decrease by 4.93%, after accounting for the effects of all unobserved variables such as time within a country and the global economy. The dynamic panel data approach establishes that around 75% of national trade openness is predetermined. In this study, the inclusion of lagged dependent variables led to a Nickell bias (Nickell, 1981) that lowered the importance of the statistic on the GPR. This paper examines the difficulties of detecting this type of bias, distinguishes it from dynamic bias within the statistical approach (Klosin, 2024), and draws macroeconomic policies from the results.

Theoretical Framework and Transmission Channels

The impact of geopolitical instability on economic integration operates through three primary, interacting transmission channels that alter a country's trade-to-GDP profile.

Investment Chilling Effect

Due to high risk premiums, not only foreign direct investment (FDI) but also domestic investment needed for sustaining and growing trade infrastructure becomes more difficult. This is because international trade demands heavy investments that must be tailored to the task, such as ports, custom-built logistics networks, and legal systems. As local geopolitical risk rises, the option value of waiting becomes greater (Caldara & Iacoviello, 2022). This means that multinational firms postpone or abandon their investments in trade infrastructure.

Decoupling in Supply Chain

Multinational companies mitigate political risks through the transition from cost reduction frameworks, like "just-in-time," to risk management frameworks, such as "just-in-case." Risk internalization leads to restructuring tactics like near-shoring, friend-shoring, and complete reshoring of factories. As local geopolitical risks increase, multinationals begin to diversify suppliers from high-risk countries; this is followed by decoupling that results in the decrease of trade/GDP ratio for the risky country (Mulabdic & Yotov, 2025).

Structural Path-Dependency

An important econometric issue when unpacking these pathways is that of structural path dependency. International business transactions rely on long-term agreements, huge investments, and well-established value chains in the global economy (GVCs). This leads to very high switching costs and inertia in the structure of international business dealings, such that current international business activity is very much dependent on past levels. An econometric model needs to differentiate between autoregressive path dependency and real contemporary geopolitical changes.

Data Construction and Variable Specification

The elaboration of the models is based on an unbalanced longitudinal panel dataset that was made by the combination of 2 main sources: The first one is the Geopolitical Risk (GPR) index collected from Caldara and Lacoviello (Caldara et al., 2026), and the second one is the Inflation and GDP growth collected from the World Bank Development Indicators (World Bank, 2026).

The variables for the models were defined with the following concepts:

- Y_{it} is the dependent variable and is defined as the trade openness, which is constructed as the sum of the nominal exports and imports of goods and services expressed as a % of the GDP, with data collected from the World Bank World Development Indicators (WDI) database.
- X_{it} is the primary independent, defined as geopolitical risk, and is derived from the GPR index constructed by Caldara and Lacoviello (Caldara & Iacoviello, 2022). The index, as specified by the creators, uses automated text-search queries to count the monthly frequency of articles related to adverse geopolitical events and threats in 10 leading English-language newspapers, normalized to a baseline value of 100.

For the control variables, as stated before, 2 main control variables were established, both derived from the WDI database:

- gdp_growth: The annual percentage growth rate of real GDP.
- Inflation: The annual CPI rate.

Econometric Specification and Empirical Strategy

3 Models were implemented and developed, which are the following:

1. Pooled Ordinary Least Squares Baseline

The original pooled OLS model without the FE and only the control groups:

$$Y_{it} = \beta_0 + \beta_1 gpr_{it} + \beta_2 gdp\_growth_{it} + \beta_3 inflation_{it} + \varepsilon_{it}$$

ε_{it} stands for the classical idiosyncratic error. Taking the assumption by the model that the intercept and slope coefficients are homogeneous across all the individual observations of the sample, which are the countries in this case.

2. The Two-Way Fixed Effects Preferred Static Model

In the second one, we now add the FE to the model, which can be defined as the time-invariant variables:

$$Y_{it} = \beta_0 + \beta_1 gpr_{it} + \beta_2 gdp\_growth_{it} + \beta_3 inflation_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

The two new variables are the α_i as the country fixed effects and δ_i as the year fixed effects.

3. Dynamic Panel Data Robustness

Lastly, to determine the long-run cumulative effect of the variables, a Dynamic Panel Data model was defined:

$$Y_{it} = \rho Y_{i,t-1} + \beta_1 gpr_{it} + \beta_2 gdp\_growth_{it} + \beta_3 inflation_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

Where $Y_{i,t-1}$ is the first lag of trade openness, and ρ captures the autoregressive inertia of international trade integration.

Empirical Findings and Parameter Estimation

	Dependent variable: trade		
	Pooled OLS	Fixed Effects	Dynamic Panel
	(1)	(2)	(3)
Trade Openness (t-1)			0.751***
			(0.028)
Geopolitical Risk Index (GPR)	-26.671***	-4.934***	-1.341
	(3.060)	(1.712)	(1.437)
GDP Growth (annual %)	-1.202*	0.423	0.182
	(0.658)	(0.303)	(0.185)

Inflation, consumer prices (%)	-0.417***	0.051	-0.009
	(0.076)	(0.048)	(0.033)
Constant	94.824***	82.467***	20.768***
	(3.644)	(1.286)	(2.462)
Observations	614	614	572
N. of groups	42	42	42
R^2	0.056	0.051	0.596
Residual Std. Error	14.422 (df=610)	1.821 (df=555)	6.148 (df=513)
F Statistic	12.059*** (df=4; 610)	9.852*** (df=59; 555)	189.261*** (df=59; 513)
Note:	*p<0.1; **p<0.05; ***p<0.01		

Data Sources: Caldara & Iacoviello GPR Index Archive (2026); World Bank World Development Indicators (2026).

Notes: Robust standard errors are reported in parentheses. Significance levels: *p<0.1; **p<0.05; ***p<0.01.

The panel is unbalanced due to listwise deletion of missing values (e.g., Argentina 2010-2017) and institutional omissions (Taiwan).

Econometric Justification and Bias Analysis

Omitted Variable Bias and the Validity of Fixed Effects

The very large difference between the Pooled OLS estimate and the Two-Way Fixed Effects estimate of the geopolitical risk parameter is a sign of strong omitted variable bias. The reason is that in the Pooled OLS specification, the constant level of trade openness as well as the reaction to geopolitical risk is taken to be constant for all countries. Yet, the F test shows strong rejection of this assumption (p<0.01).

With the Within estimator applied to Model 2, the empirical approach effectively controls for unobservable, time-invariant heterogeneities. In the case of pooled OLS, issues like geographic isolation, cultural similarities, and institutional structure are all contained within the error term. Since these variables have high correlations with not only a country's predisposition toward engaging in trade but also its level of geopolitical risk, the Pooled OLS estimate will be grossly upwards-biased. After accounting for time-invariant characteristics (α_i) and global shocks (δ_t), the TWFE regression produces the true effect: when considering all else being equal, a unit change in domestic geopolitical risk results in a 4.93 percentage decrease in openness to trade vis-à-vis GDP.

Path Dependency and the Nickell Bias Dilemma

Model 3 highlights the structural friction embedded in the process of integrating the global supply chain. The computed estimate for the autoregressive term is extraordinarily large and extremely statistically significant (ρ = 0.7507, t = 27.103). Thus, roughly 75% of the trading integration level of any country in a particular year is predetermined by its trade architecture the previous year. This result is in line with economic reasoning, since supply chains, transportation agreements, and international shipping routes cannot be easily undone or recreated due to short-term events. More importantly, since $|\rho| < 1$, the dynamic process is stationary.

However, due to Nickell bias (Nickell, 1981) in Model 3, Model 2 is still considered more appropriate to be used as a baseline in estimating policies from theory because, in the panel data sets where time dimension ($T \approx 15$), demeaning process involved in removing country fixed effects mechanically leads to correlation between lagged dependent variable ($Y_{i,t-1}$) and idiosyncratic error term ($\widehat{\varepsilon_{it}}$). This leads to violation of exogeneity requirement, resulting in biased estimates of ρ that cannot be solved by increasing cross-section sample size (N).

Key Findings from the Regression Analysis

TWFE emerges as the most reliable static specification for capturing the pure impact of political risks. After controlling for unobserved, fixed country traits such as geographic insulation or deeply entrenched institutions as well as worldwide economic fluctuations, the TWFE regression reveals a highly informative coefficient:

An increment of one unit in the domestic Geopolitical Risk (GPR) Index causes a 4.93 percentage point decrease in the ratio between trade openness and the size of GDP (p = 0.0041).

It is therefore clear that despite adjustments for real GDP growth and stable exchange rates, political risks directly contribute to slowing down the pace of economic integration in this setting. The extreme bias found in the baseline pooled OLS estimator, which estimated the coefficient at an inflated value of -26.671, shows how omitting country heterogeneity biases the estimation by treating domestic political vulnerability as a proxy for trade patterns.

The Dominant Inertia of Trade Architecture (Dynamic Model)

The introduction of a dynamic factor brings into focus the true physical and contractual friction involved in international business dealings. The autoregressive coefficient is extremely large and statistically very significant:

$$\rho = 0.7507\ (t = 27.103, p = 0.0000)$$

What this implies is that 75.07% of the openness level of any country to business in any particular year is dependent on its structural framework of the previous year. GVCs, logistic agreements, and transport infrastructure cannot be readily disintegrated or reconfigured.

Conclusions

As can be observed, the empirical results present a stark juxtaposition between a momentary shock and intrinsic structural persistence. In specific terms, an increase in geopolitical risk in a particular country leads to a decrease in the openness of such countries' economy by 4.93 percent of GDP through serving as an unseen tariff of up to 14% in extremely volatile industries (Mulabdic & Yotov, 2025). Nevertheless, globalization proves to be extremely resilient and rigid: 75.07 percent of today's openness of economies is defined by the structure formed a year before, showing that global value chains and logistics networks cannot be redirected or friend-shored on the fly.

Such an imbalance poses severe threats to the process of risk assessment and prediction since the traditional static methods would either vastly exaggerate the speed of separation within supply chains or entirely overlook the threat due to Nickell bias effect (Nickell, 1981). The only solution lies in utilizing more advanced, dynamic estimation techniques that are capable of mitigating the effect and accounting for supply chain decoupling (Klosin, 2024). Such methods include Klosin Dynamic Bias Correction (Klosin, 2024) and Bruno's LSDVC estimators (Bruno, 2005), among others.

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